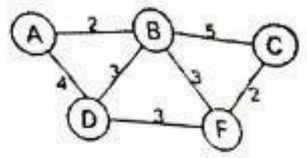
University of Barishal

Dept. of Computer Science and Engineering

Course Code: CSE-3205:; Course Title: Artificial Intelligence

Admission Session: 2020-21; **Exam:** Mid Term-II; **Marks:** 20; **Time:** 50 min



1. Perform `Breadth-First Search (BFS)` starting from node **A**. List the order in which nodes are visited. 5

2. Using *A* algorithm*, find the shortest path from **A** to **F**. Assume the heuristic (straight-line distance to F) is: 5

h(A)=5, h(B)=4, h(C)=2, h(D)=6, h(E)=3, h(F)=0

Show your step-by-step calculation, including f(n)=g(n)+h(n).

3. Draw the final search tree showing the paths explored by A*. 5

4. Explain which algorithm (BFS or A*) is more efficient in this case and why. 5

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